

What is Radio Frequency?

Radio frequency (RF) refers to a range of frequencies within the radio spectrum, which is part of the electromagnetic spectrum. The frequencies range from 3kHz to 300 GHz. These radio waves are the backbone of modern technologies such as telecommunication devices, navigation signals, wireless networking, and radio broadcasting.

What are “Radio Frequency Parameters”?

Radio Frequency Parameters are the parameters used to define, explain, and explore radio frequency concepts. On this poster, the introductory parameters such as impedance (Z) , admittance (Y) , transmission line (ABCD), and scattering (S) parameters will be covered.

I. Z Parameters

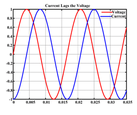
a. **Impedance** is the resistance of flow in alternating current (AC) . Direct current (DC) has one real value of resistance, notably named resistance, and is represented by R, which is measured in Ohms (Ω). This leads to Ohms Law, where V represents voltage, and I represents current [1]. However, when considering alternating currents, both voltage and current change alongside time. This rebuilds Ohms law [2].

$$R = \frac{V}{I} \quad [1] \qquad Z = \frac{V}{I} \quad [2]$$

Z represents impedance, more formally known as **complex impedance** [3]. The term complex is derived from Z being a combination of a real component, resistance, and a reactive component X, represented mathematically as jX. These components represent the **magnitude** and **phase** relationship between the current and voltage. A **positive reactionary component** [3a] indicates **inductive impedance**, and a **negative reactionary component** [3b] indicates a **capacitive impedance**. The magnitude of Z is the length of the vector, and shows how much current the circuit resists. Phase represents when the current flows relative to voltage. A positive phase angle indicates inductive impedance, and a negative phase angle indicates capacitive impedance. This is calculated with a formula [3c]. In an alternating current, inductive impedance [3a] is when current lags behind voltage, due to resistors delaying the change in current. Capacitive impedance [3b] occurs when current leads voltage, since capacitors resist change in voltage.

Inductive Impedance

$$Z = R + jX$$

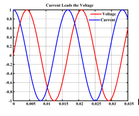


[3a]

$$\phi = \tan^{-1}\left(\frac{X}{R}\right) \quad [3c]$$

Capacitive Impedance

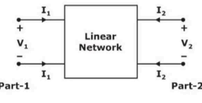
$$Z = R - jX$$



[3b]

b. For a single port network, this would be enough. However, most RF networks molded as multiport networks. To characterize the interactions between voltage and current’s at each port, impedance parameters are utilized, where each Z component represents complex impedance.

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \quad [4]$$



[5]

A port is a pair of terminals where current enters and exits. In this case, port 1 is represented by V_1 and I_1, and port 2 is represented by V_2 and I_2. Z parameters describe the relationship between voltage and current of these two ports, capturing how a signal at one port effects both ports.

$$V_1 = Z_{11} * I_1 + Z_{12} * I_2$$
$$V_2 = Z_{21} * I_1 + Z_{22} * I_2 \quad [6]$$

Z_{11} and Z_{22} both represent input and output impedances, respectively, whereas Z_{12} and Z_{21} explain how current at port 2 affects the voltage at port 1, and how current at port one affects the voltage at port 2, respectively. The equation to solve for each individual Z parameter is as followed:

$$Z_{nm} = \frac{V_n}{I_m} \Bigg|_{I_k = 0,\ k \neq m} \quad [7]$$

When port 1 is **open circuited** ($I_1 = 0$) & ($R_X = \infty$), Z_{12} and Z_{22} can be solved for. An open circuit is a condition where an electrical path does not flow. The terminals are disconnected, meaning continuous flow between both ports is not happening. Although current is prevented from flowing, voltage still exists. When port 2 is open circuited ($I_2 = 0$), Z_{11} and Z_{21} can be solved for.

c. Radio frequency components and networks rely on the relationship between current and voltage. These parameters capture how signals are transmitted, absorbed, and reflected. When $Z_{12} = Z21$, networks (devices with one or more ports) are **reciprocal**. The transfer of signal in a reciprocal network is the same from both ports. For a network to be reciprocal, it must be also passive. Coaxial cables [8] are an example of a passive network; it does not modify or add energy to the signal. When $Z_{11} = Z_{22}$, the network is symmetrical. You can swap the input and output ports and get the same result. All symmetrical networks are reciprocal, but the reverse is not true. In RF design, symmetry matters for matching impedance, which will be expanded upon later. Z parameters are best used in DC or low kHz range in a RF context.



[8]

II. Y Parameters

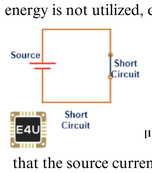
Y parameters are the inverse of Z parameters. It is measured in siemens (S), and explains how easy current flows through a network [9].

$$Y = \frac{I}{V} \qquad Y = \frac{1}{Z} \quad [9]$$

Y parameters are also constructed similarly to Z parameters. The distinct difference is that current and voltage are flipped [10].

$$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} \quad [10]$$

Y parameters are known as “short circuit admittance parameters,” and are calculated under short-circuit conditions. Essentially, $I_X = \infty$, and $V_X = 0$. In short-circuit conditions, resistance is ideally zero, making I ideally infinite. **Short circuits** [11] essentially are a loop where energy is not utilized, due to no components that use energy being present. Since there is no impedance, there is no voltage drop. Voltage in itself is the potential difference between two points; no impedance means that the source current is unchanged. That is why voltage is 0, due to the lack of change. The admittance parameters [12] represent current in terms of voltage, and the equations themselves are similarly flipped.



[11]

$$I_1 = Y_{11} * I_1 + Y_{12} * I_2$$
$$I_2 = Y_{21} * I_1 + Y_{22} * I_2 \quad [12]$$

Y_{11} and Y_{22} both represent short circuit input and output admittances, respectively. Y_{12} and Y_{21} are short circuit **transfer admittances** from the input to the output port and short circuit transfer admittances from the output to input port, respectively. Transfer admittances refer to the relationship between voltage at one point and current at another. Y parameters are best used in DC or low kHz range in a RF context.

III. ABCD Parameters

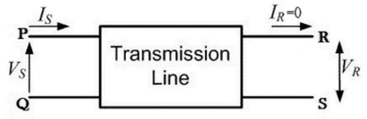
a. ABCD parameters primary purpose are for **cascading networks**. An example of this are transmission lines [13]. Transmissions lines are conductors that carry large volumes of electrical power. **Transmission matrices [14]** are useful due to how common two-port networks are.



[13]

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \quad [14]$$

Examples of one port networks in the context of RF are antennas and dummy loads (devices that simulate electrical loads). Two port networks are components such as filters and amplifiers.



[15]

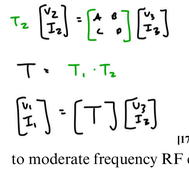
The ABCD parameters vary depending on if calculations are being made on an open circuit (I = 0) or a short circuit (V = 0). On an *open circuit*, *A* is the **voltage ratio**, and *C* is open circuit **conductance (Ω)**. The voltage ratio is the relationship between input and output voltage in a transformer and is expressed as a fraction. Conductance is the capacity of a component to allow the passage of a current, and is the opposite of resistance. On a *short circuit,* B represents **short circuit resistance** (mho), and D represents the **current ratio**. Similar to open circuit conductance, B represents the resistance of the transmission line. D is the ratio of the initial current to the ending current.

$$A = \frac{V_S}{V_R} \qquad B = \frac{V_S}{I_R}$$
$$C = \frac{I_S}{V_R} \qquad D = \frac{I_S}{I_R} \quad [16]$$

To calculate A and C [16], $I_R = 0$, otherwise an open circuit. To calculate B and D, $V_R = 0$, otherwise known as a short circuit.

b. What are the use cases of ABCD parameters?

One prominent use case is cascading. Cascading is when you have a line of two-port networks, each with voltages and currents starting with V_1 ,I_1, and ending with V_N ,I_N. In between, there are *N* numbers of *T* [17]. Additionally, symmetrical circuits can be determined when A = D. Circuits are reciprocal when the determinant of the ABCD parameters is equal to one. ABCD parameters are also heavily dependent on the length of a transmission line. ABCD parameters are best used in low to moderate frequency RF circuits. The higher a frequency is, the harder it is to measure voltage and current, making it harder to calculate these parameters.



[17]

IV. S Parameters

a. What are S Parameters? S parameters, also known as scattering parameters, are used to represent a network’s radio frequency characteristics. They are critical for depicting how much of a signal is reflected back, transmitted through, or transferred between points. S parameters, similar to Z and Y parameters, have a magnitude and phase component. The magnitude represents how much of the signal is transmitted or reflected, while phase indicates what the timing of the signal is. S parameters also do not use open or short circuit conditions to complete calculations.

$$\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \quad [18]$$

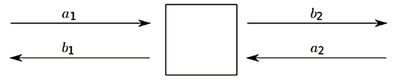
S parameters [18] have a specific naming convention. The first subscript denotes output port, and the second subscript indicates input port. For example, if a signal enters from port one and exits from port two, the parameter is denoted as S_{21}.

Parameter	Name	Meaning
S11	Input Reflection Coefficient	How much of the input signal is reflected due to mismatched impedances.
S21	Forward Transmission Coefficient	The efficiency of the signal transmission from the input port to output port.
S12	Reverse Transmission Coefficient	How well port 1 is isolated from signals entering port 2.
S22	Output Reflection Coefficient	How much of the output signal is reflected due to mismatched impedances at output port.

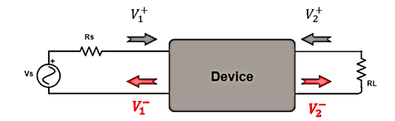
[18]

b. What are reflection coefficients?

To understand reflection coefficients, first **incident** (a1, a2) [19a] **waves** and **reflected** (b1, b2) [19a] **waves** need to be defined. When a signal is applied to a port, it splits into two waves: an incident wave, and a reflected wave [19b].



[19a]



[19b]

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad [19c]$$

V_1^+ represents the incident wave and V_1^- represents the reflected wave at port 1. V_2^+ represents the incident wave and V_2^- represents the reflected wave at port 2 [19]. An incident wave is simply a wave that travels through a source; a reflected wave is a wave that encounters some sort of “boundary” and reflects back. An example of this could be a mismatched impedance.

$$S_{11} = \frac{V_1^-}{V_1^+}|V_2^+ = 0 \qquad S_{12} = \frac{V_1^-}{V_2^+}|V_1^+ = 0$$
$$S_{22} = \frac{V_2^-}{V_2^+}|V_1^+ = 0 \qquad S_{21} = \frac{V_2^-}{V_1^+}|V_1^+ = 0 \quad [20]$$

S_{11} [20] is the ratio between the reflected wave and incident wave at port 1 when the output is loaded with a matched load (V2+ = 0) (otherwise known as the **input reflection coefficient**). A lower S11 value is better, as it essentially that means that the signal is being absorbed to the network rather than reflected back. S_{12} [20] is the ratio of the reflected wave to the incident wave, when V_1^+ is zero. This essentially quantifies the reverse isolation of the circuit, depicting how well the output is isolated from the input. Higher values of S_{12} indicate better isolation between input and output ports. S_{22} [20] measures how accurate the outputs match (otherwise known as the **output reflection coefficient**). Similar to S_{11}, a lower S_{22} indicates better output matching, and power transfer is optimized. The input port is terminated with a matched load, and a low S_{22} indicates most of the signal is being absorbed by the load. S_{21} [20] is the **gain** of the circuit, which essentially measures how effectively the circuit amplifies the input signal. The higher S_{21} is, the larger portion of output signal that is being fed back to the input network; this is crucial in amplifier designs, where signal amplification is the primary goal.

c. What are the use cases of S Parameters?

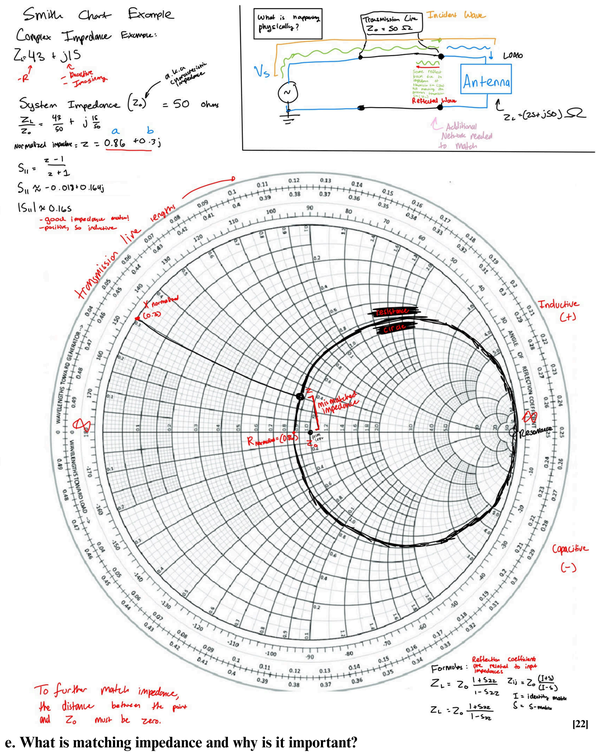
S parameters excel at measuring high moderate to high frequencies, due to the fact that incident and reflected waves are analyzed. The parameters are analyzed by a vector network analyzer (VNA) [21] and provide information such as information on the incident and reflected waves. These parameters and waves also fold into **Smith Charts**.



[21]

d. What are Smith Charts?

Smith charts [22] are a graphical tool that help visualize impedances, and it has a particular strong suit in measuring mismatched impedances. They can also be used to measure admittances (Y parameters), and even “imittances” (a mix between impedance and admittance parameters).



[22]

e. What is matching impedance and why is it important?

Matching impedances is a critical concept in RF theory, especially when considering transmission lines. When impedances are mismatched, signals are much more likely to be reflected, causing data loss. This essentially means the source and load do not have the same impedances. In our example [22], the source (Vs) and load (Antenna) did not match, meaning impedance was mismatched. Another example of this is **lossy** vs **lossless** audio files. An audio file is theoretically lossless when the magnitude of S_{11} and S_{21} squared is equal to one [22a]. If it is below 1, it is lossy [22b].

$$|S_{22}|^2 + |S_{21}|^2 = 1 \quad [22a] \qquad |S_{22}|^2 + |S_{21}|^2 < 1 \quad [22b]$$

In the lossy audio file, some of the audio is discarded, making it sound less in tune to the original song. Whereas, if you select the lossy option when listening to music, it sounds higher quality, as the original data of the song is preserved. Matching impedances are the backbone to signals. RF design focuses a lot of matching networks, which is essentially the process of combining components until impedances match. An example of a matching network would be a network that goes inbetweener an amplifier and antenna with different impedances. It leads to optimal power transfer and low data loss.



Bibliography... powered by your local cascaded radio frequency network in your hands!